

Car Rental Management System

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Introduction

Project Overview

The Car Rental Management System is a web application that helps people rent vehicles easily. It is like a bridge between people who own car rental businesses and people who want to rent cars. The Car Rental Management System lets people look at the cars they can rent make a booking and pay for it online in a way. At the time it gives the people in charge of the car rental businesses some useful tools. These tools help them keep track of the cars manage the bookings and get reports on how the business doing at that moment. The Car Rental Management System is about making the car rental process easier, for the Car Rental Management System users.

1.2Background

- The car rental business has always done things the way. They used paper to keep track of things. People had to go to the office to book a car. They had to call to ask questions.
- Things are changing fast with all the new technology in delivery and ride services like Uber and Careem.
- Now people want things to happen away. They want to be able to get what they need instantly.
- The car rental business needs to change. The car rental business needs to start using technology if they want to stay in the car rental business and keep running.

- The car rental business has to keep up with the times and stay competitive, with car rental businesses.

1.3 Literature Review / Similar Systems

Modern day services like Uber and Careem have established an exceptionally high standard in real-time logistics and the end-user experience. By providing a platform that offers real-time availability of rentals with little to no latency, as well as pricing that is fair and transparent; this system uses these benchmarks as the foundation for building a new and better method of booking cars that relies on electronic rather than manual systems, therefore reducing the likelihood of human error in booking schedules and in maintaining proper financial reporting with regards to bookings.

The car rental system we have now is mostly done on paper. People who want to rent a car have to go to the office or call them to see if a car is available. The problem with this system is that it does not have a place to store all the information. This means that customer information the cars we have and money transactions are all done by hand and not updated in time.

1. Pak car rentals
2. New Pakistan motors

1.4 Purpose

The purpose of this Software Requirements Specification (SRS) is to define a well-organized, detailed, and easy-to-understand set of guidelines for the development of the Car Rental Management System. This document includes all

functional and non-functional requirements needed to automate the car rental process. As a result, this SRS will serve as a formal agreement between the developers and other involved parties to assist in the creation of a sound architecture for a modern car rental business

1.5 Issues and Motivation

Issues:

The current system is prone to mistakes because it is done by hand. For example someone might book a car that is already taken. Also customers do not have access to information about prices. If a car is available right away. The people in charge have a time keeping track of money because the system is not efficient.

Motivation:

We want to create a system to fix these problems. The new system will be fast and fair. It will make sure that customers know what they are getting. Our goal is to make a system that's electronic reduces mistakes and makes the experience better for the people who use it. We want car rentals to be easier and more straightforward for everyone, including the car rental company and the customers, like the people who want to rent a car. The car rental system will be better because it will have a place to store information and it will be updated in real time which means the car rental company and the customers will have the same information at the same time.

1.6 Problem Statement

The car rental businesses do not have a system to manage their daily work. This is a problem. For example there are delays and mistakes with bookings. The customers are not happy with the car rental businesses. We need a system that makes it easy for customers to find cars, book cars and pay for cars online. At the time the car rental businesses need a strong tool to manage the cars track the rentals and handle the money safely. The car rental system should be easy to use for the customers and the car rental businesses.

1.7 Scope

- The Car Rental Management System is a centralized web application that will be used to manage the relationships between car rental companies and their customers.
- The Customer Module (1) allows the customers to: register an account; search for vehicles by filter (brand, model, price range); book vehicles; and automate billing.
- The Admin Module (2) provides the administrator (Owner) with tools to: manage their fleet; manage their customers' accounts; track the status of the fleet's bookings; and generate usage and profit reports.

1.8 Project Management / Project Management Plan

We will manage this project. Our team will use tools like Jira or Trello to break the project into tasks. We will finish these tasks one by one we call these tasks sprints. We will have meetings to check how we are doing with the car rental system. We want to see if there are any problems with the car rental system and make sure we are meeting all the requirements for the car rental system. We will use version

control like Git or GitHub to work on the code for the car rental system. This will help us keep track of the changes we make to the car rental system.

1.9 Process Model

We will use the Agile Software Development Life Cycle model for this project. We will use the Scrum framework for the car rental system. We chose this model because it lets us develop the car rental system in parts. We will get feedback all the time on the car rental system. We will start with the Customer Module of the car rental system. Then we will work on the Admin Module of the car rental system. This way we can make changes to the car rental system if we need to. We want to make sure the final car rental system is really good. We will develop the car rental system in parts. We will make sure each part of the car rental system is working well before we move on to the part. This will help us make a high-quality car rental system that meets the needs of the car rental businesses and their customers. The car rental system will be easy to use and effective, for the car rental businesses and their customers.

1.10 Product Perspective

This solution replaces the previous paper systems with a new digital solution. The Car Rental Management System is a 3-layered, web-based solution with a front-end, a back-end, and a database. The solution uses the internet GIS API and internet credit card processing real-time APIs.

1.11 User Roles

User Role Description of Guest:

Guest - A person who has not yet logged into the System through the client website Registered Customers Registered users on the system. Users can manage their account profile, reserve a vehicle, give ratings and feedback, and make payments using a secure payment method.

Administrators / Operators

Authorized personnel who are responsible for maintaining the car fleet and monitoring the overall health of the rental system by approving rental requests in the system.

1.12 Proposed Technologies

- Frontend: HTML5, CSS3, (Bootstrap), JavaScript
- Backend: Node.js/PHP (server-side language)
- Database: MySQL (for structured storage of relational data)
- APIs: Google Maps API for tracking and Stripe/PayPal for processing payments

2. SRS

2.1 Requirements

2.1.1 Functional Requirements:

- ID Title Detailed Functional Requirement
- **FR001** Registration/User Creation The system must allow for the user to register as a user by entering a name, email address, 11-digit phone number, and creating a secure password.
- **FR002** Search Vehicles for Rent The user must be able to search the available fleet (cars to rent), including, but not limited to location, date range, and vehicle category.
- **FR003** Calculate Reservation Cost Reservation cost must automatically calculate as the user enters a temporary rental agreement and reserve the vehicle they are requesting to rent.

- **FR1.4**: The system shall allow customers to view and update their personal information after logging in.
- **VALIDATION** through databases and through nadra for age and from authorities like is the candidate have clear record.

2.1.2Non-functional requirements:

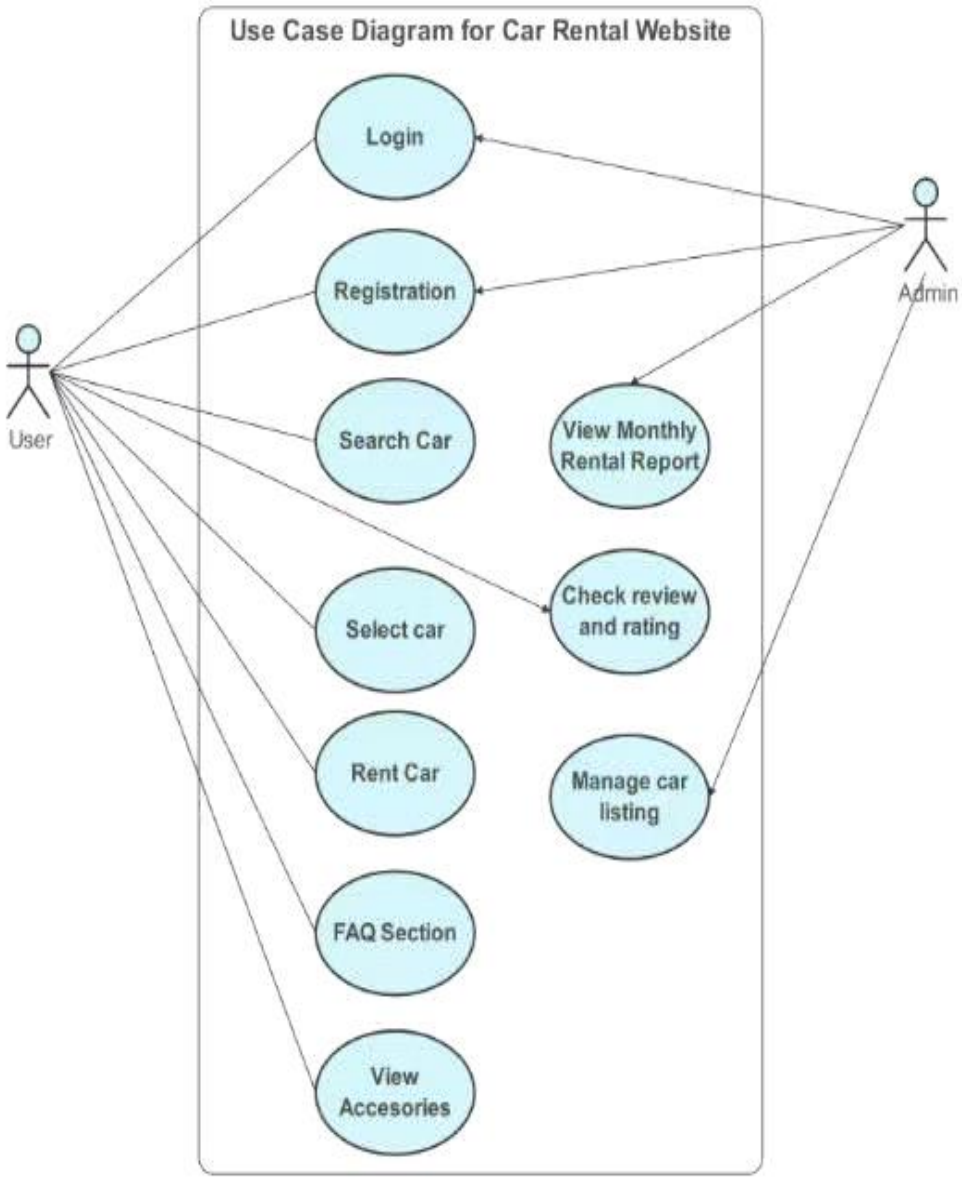
An overview of non-functional requirements below will give you a general idea of what is required for this application.

- Performance: Search queries should return results within 1.5 seconds at most.
- Scalability: The system architecture must be able to adapt to the increase of vehicles and users while maintaining a level performance.
- Security: All sensitive information (i.e., password, personal identification) must be encrypted using SHA-256 or equivalent.
- Availability: The system should provide 99.9% availability for users to be able to

reserve a vehicle at any time.

- Once the application is developed, it will have specific processes or use cases for the application to follow. The example of the Use Case below is for the sign-up process. Each process or use case will display the actor(s), preconditions, basic flow, and exceptions to ensure users can book vehicles at any time.

2.2 Use Cases and Flow of Processes



Use Case : Sign Up (UC001)

Use Case Description Doc: Sign Up

Name of Use Case: Sign Up / User Registration

Use Case ID / No.: UC – 001

Use Case Purpose

The Car Rental System Sign Up process is designed to allow a guest user to create an account. This is the goal of the Sign Up process. The Car Rental System Sign Up process is very important.

Actors

- The main actor is the Guest, who's an unregistered user.
- The Secondary Actors are the Database and the System.

Preconditions

To start the Sign Up process the user needs to have a working internet connection.

- The user must not already have an account with the email address they want to use for the Car Rental System.

Postconditions:

- After the Sign Up process is complete a new user profile is recorded in the

database.

- The user can now log into the Car Rental System using their account.

Scenarios (Flow of Events)

Basic Success Scenario (Happy Path)

Steps:

1. The guest user clicks on the Sign Up or Register button on the Car Rental System homepage.

2. The Car Rental System shows the registration form to the guest user.

3. The guest user enters their details, including Name, Email address, phone number and a secure password into the registration form.

4. The guest user submits the registration form to the Car Rental System.

5. The Car Rental System checks the entered information to make sure it is correct.

6. The Car Rental System saves the user details in the Customer Database.

7. The Car Rental System shows a registration success message to the guest user.

8. The guest user is redirected to the Login page of the Car Rental System.

Alternative Flow (Extensions)

* A1 – Missing Information: If the guest user submits the form without filling in all the fields the Car Rental System will highlight the missing fields. Ask the guest user to complete them.

Failure scenarios:

* F1 – Email Already Exists: If the guest user enters an email address that is already registered with the Car Rental System the Car Rental System will reject the request and show an error message. The guest user is prompted to either log in or use an email address.

* F2 – Invalid Phone Number: If the guest user enters a phone number that's not 11 digits the Car Rental System will show an error message. The guest user is asked to enter a phone number.

* F3 – Weak Password: If the entered password does not meet the Car Rental System security criteria the Car Rental System will ask the guest user to create a password.

Other Elements

Triggers

A guest user wants to rent a vehicle from the Car Rental System and needs to create an account to do so.

Assumptions

* The guest user has an email address that they can use to create an account, with the Car Rental System.

* The Car Rental System database is online and accepting entries.

Use Case : renting car (UC002)

Use Case Description Doc: Renting a Vehicle

Name of Use Case:Renting A Vehicle.

Use Case ID / No.::UC – 01

Use Case Purpose

To allow customers to search for, select and rent a car through the Car Rental System.

Actors

Primary actor

Customer

Secondary Actors

Payment Gateway

Admin

Car Inventory System

Preconditions

- The customer must have a registered account or provide valid information to register an account.
- There must be vehicles listed in the database.
- The customer must have an internet connection.

Postconditions

- The rental reservation is recorded in the database.
- The payment has been processed.
- The selected vehicle's status changes to "Rented".
- The customer will receive a rental confirmation by e-mail.

2. Scenarios (Flow of Events)

Basic Success Scenario (Happy Path)

Steps:

1. Customer logs into the Car Rental System.

2. Customer searches for available vehicles for rent based on location, rental date and type of vehicle.

3. The Car Rental system returns a list of vehicles available for rent.

4. Customer selects an available vehicle.

5. The Car Rental System displays the details (Rental Rate & Terms) of the vehicle selected.

6. Customer enters their personal & Drivers License information.

7. Customer confirms the details of their rental reservation.

8. The Car Rental System calculates the total rental fee.

9. The customer is given the option to choose how they would like to pay for the

rental.

10. The Payment Gateway processes the payment.

11. The system confirms that payment has been successfully completed.

12. The system generates and provides the customer with their Rental Booking ID.

13. An email is sent to confirm the booking.

Alternative Flow (Extensions)

1. **A1 – Customer Uses Alternate Mode of Payment:** Customers who choose to pay using alternate methods (cash, credit card, or online wallet) will be redirected to their selected method of payment and instructed to finish processing their payment in order to finish making their reservation.
2. **A2 – Customer Updates Pick-Up/Drop-Off Dates:** When a customer updates the pick-up and/or drop-off date(s) of an existing reservation, the rental price will be automatically calculated according to the newly requested pick-up and drop-off dates.
3. **A3 – Customer Enters Valid Coupon Code:** If a customer enters a valid coupon code at the time of check-out, the value of the coupon will be deducted from the amount due.

Failures scenarios:

- **F1 – Vehicle Not Available**: The system will continually monitor vehicle availability before continuing with the reservation process; if a vehicle has been reserved by another customer, the reservation will not be allowed to continue.
- **F2 – Transaction Failed**: In the event that a transaction fails for any reason, (e.g. due to interference with customer's payment processor or issues with the customer's credit/debit card) the reservation will be cancelled and must await payment resolution.

F3 : Invalid Driving License

1. Customer enters invalid or expired license information.

2. System rejects the request.

3. Customer is asked to provide valid information.

Other Elements

Triggers

The customer wishes to hire a car and begins the booking process.

Assumptions

1. The customer holds a valid driving licence.

2. The cars available through the system will be appropriately maintained.
3. The payment systems are functioning.

Business rules

- You must be 18 years or older in order to rent a car.
- You must also possess a valid driver's licence in order to rent a car.
- You must pay for your rental prior to picking up the vehicle.
- Rentals may not exceed 30 days unless prior approval has been received.
- If you do not return the vehicle on time, you may incur additional charges.

2.3 Use Case Model

- We noticed that this section was not in your document so we created one that fits the use cases in your document.

- The Car Rental System Use Case Model shows how people and things interact with the Car Rental System.

- The Car Rental System Use Case Model is like a picture that explains how everything in the Car Rental System works together.

- There are people and things that use the Car Rental System.

- These people and things are called Actors of the Car Rental System.

- The Car Rental System Actors are:

- Customer: The Customer uses the Car Rental System to sign up look for cars and book a rental with the Car Rental System.

- Admin: The Admin uses the Car Rental System to manage the cars and oversee the bookings in the Car Rental System.

- Payment Gateway: The Payment Gateway is like a helper that makes sure the money moves from one place to another for the Car Rental System.

- Now let us talk about how these Car Rental System Actors work with each other in the Car Rental System.

- The main things the Car Rental System Actors do are called Use Cases of the Car Rental System.

- Here are some important Car Rental System Use Cases:

- When you Make A Reservation with the Car Rental System you also have to Process A Payment with the Car Rental System.

- You cannot finish making a reservation with the Car Rental System without paying for the Car Rental System.

- When you Make A Reservation with the Car Rental System you can also Apply A Coupon with the Car Rental System if you have one for the Car Rental System.

- This is like a step that you can do if you want to with the Car Rental System.

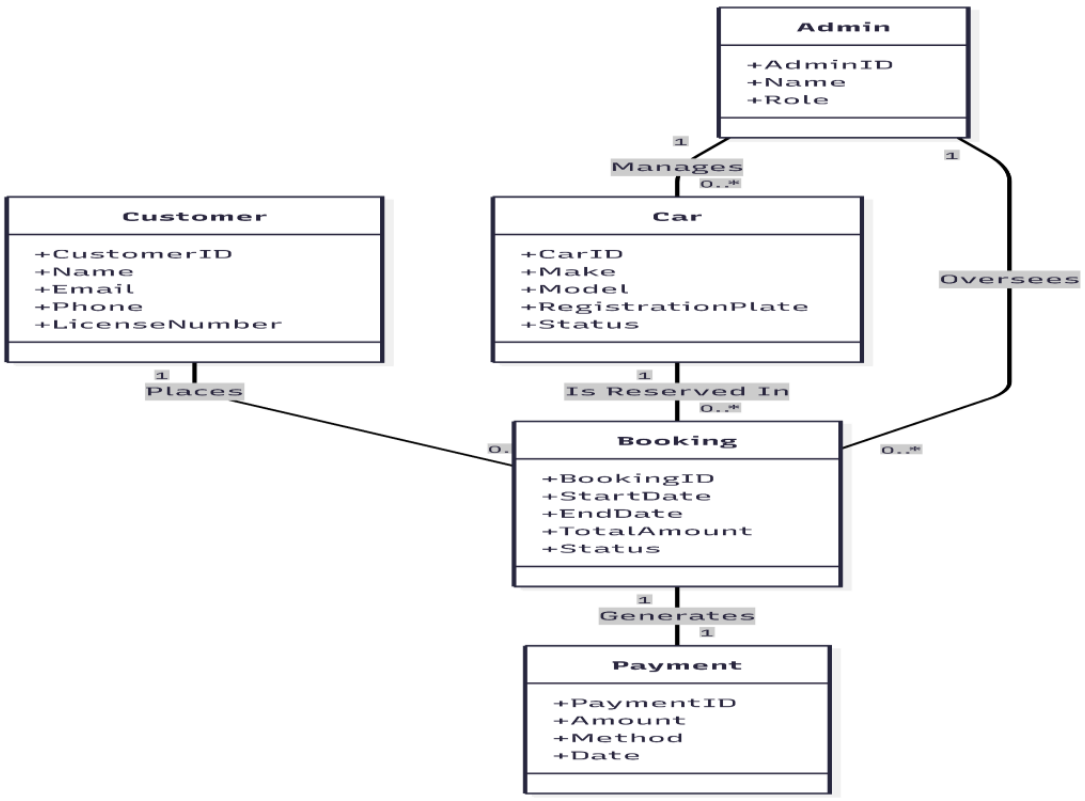
- You have to Sign Up or Login to use the Car Rental System.

- This is so Customers and Admins of the Car Rental System can get to their pages, in the Car Rental System.

- The Car Rental System needs to know who you are before you can use the Car Rental System.

ANALYSIS:

Domain model:

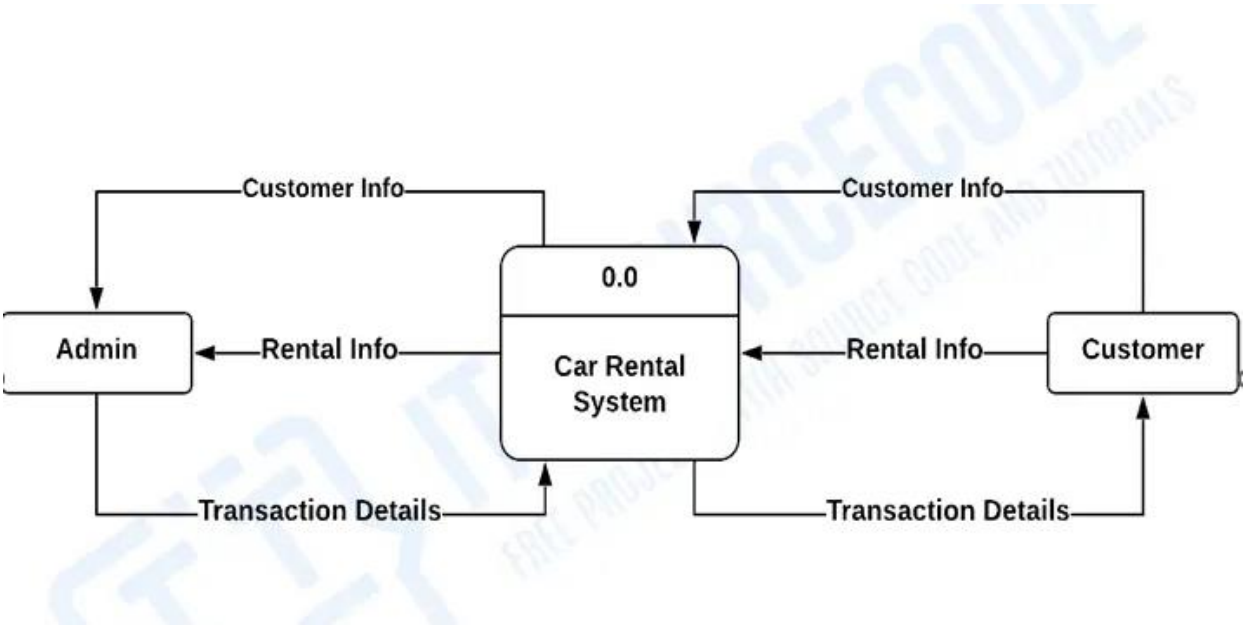


Data flow diagram:

Car Rental System DFD Level 0

The DFD Level 0 is also know as the context diagram of Car Rental System. Car Rental System context diagram presents the main idea to be the basis for the proceeding levels. The basic idea is represented by a single process consisting the main process, users and data

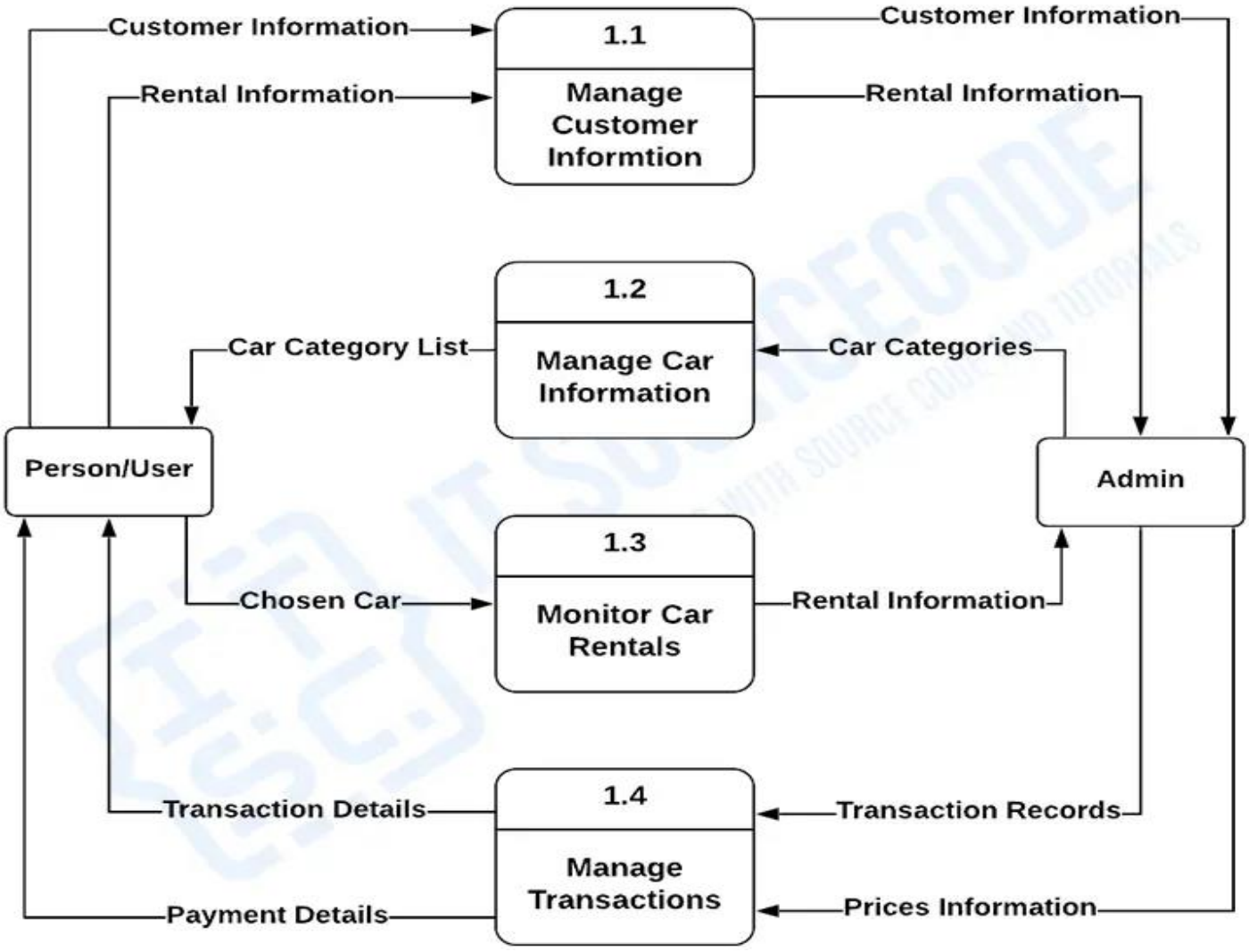
This context diagram portrays the general process done in Car Rental System. The general process is based on the concept of the proposed project and illustrated using the DFD symbols to visualize the idea. This will also serve as a guide as you go through the deeper processes of the car rental system data flow diagrams.



Car Rental System DFD Level 1

Next to the context diagram is the level 1 data flow diagram. It contains the broadened

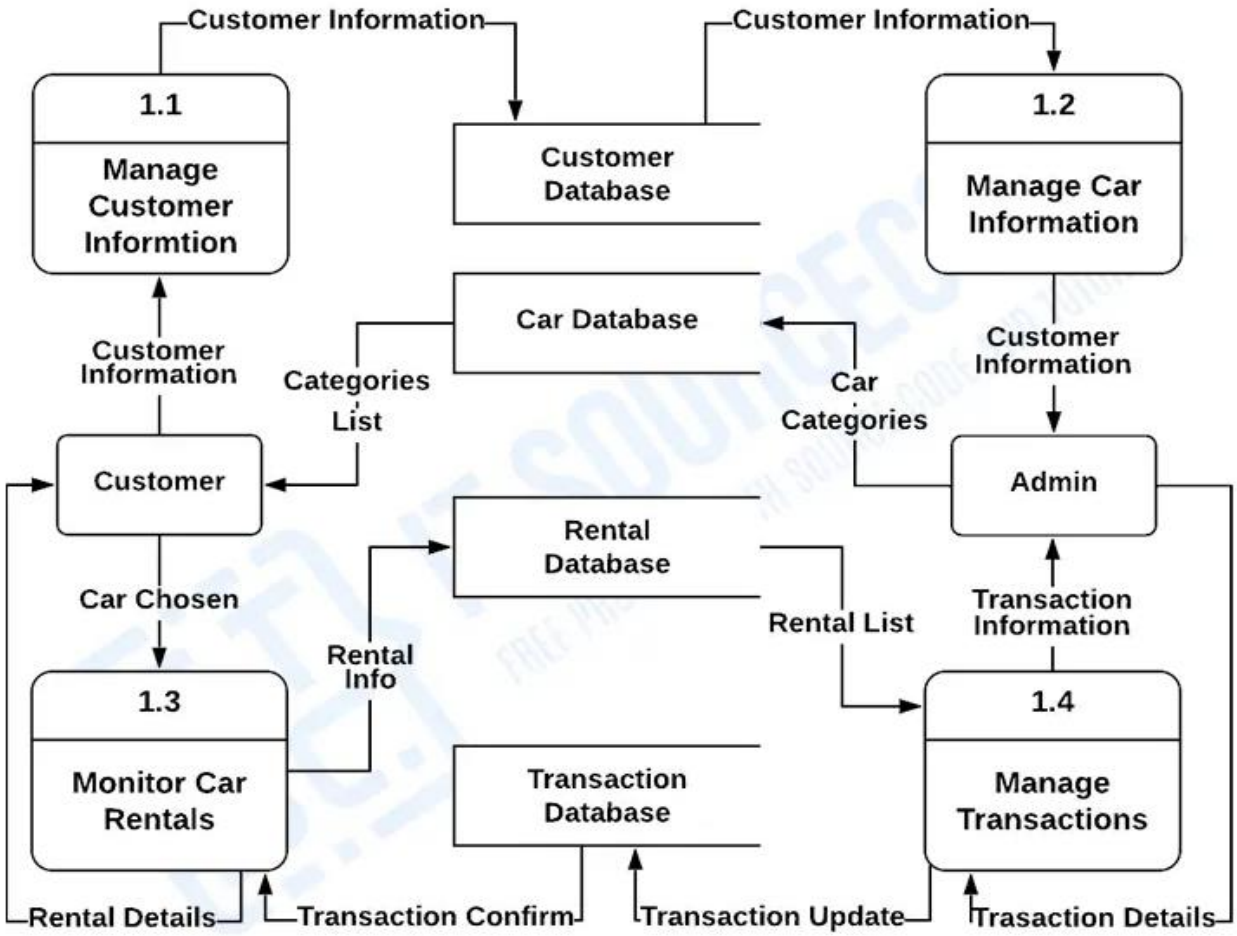
concept of DFD Level 0. It'll be done by determining the included processes to complete the basic idea from the context diagram. So as the processes were determined, data inputs will be specified as well as the outputs.When the processes and data were all specified, you just have to plot the flow of data from the external entity (user) to the process up tp producing the output. With being knowledgeable about the DFD level 1 of the Car Rental System, you will know then its broaden context terms.



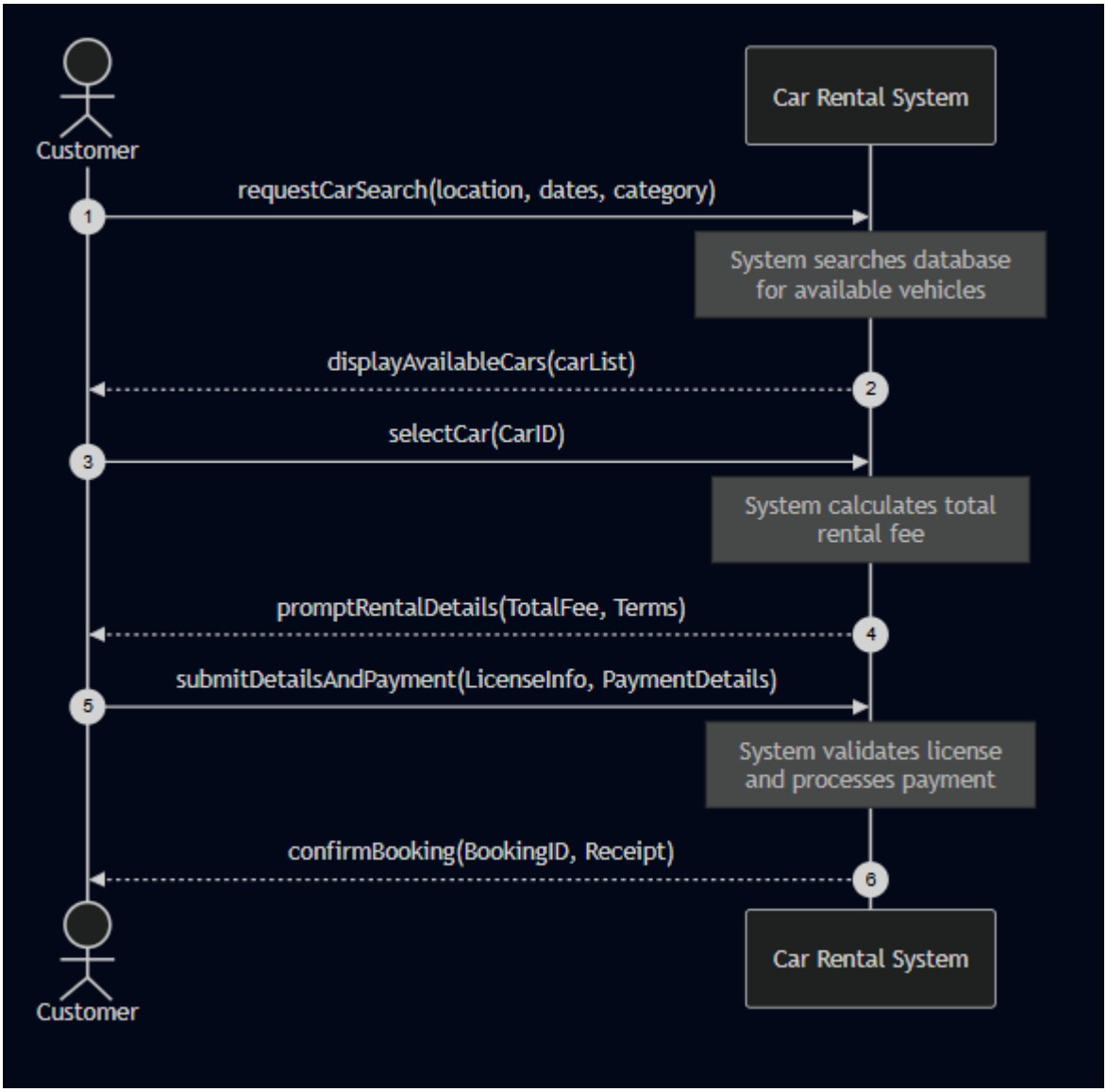
Car Rental System dfd Level 2

After presenting the Car Rental System DFD levels 0 and 1, next to that is level 2. The DFD Level 2 was considered as the highest abstraction of Car Rental System Data Flow Diagram. This level is expected to have the complete and detailed illustration of the project.

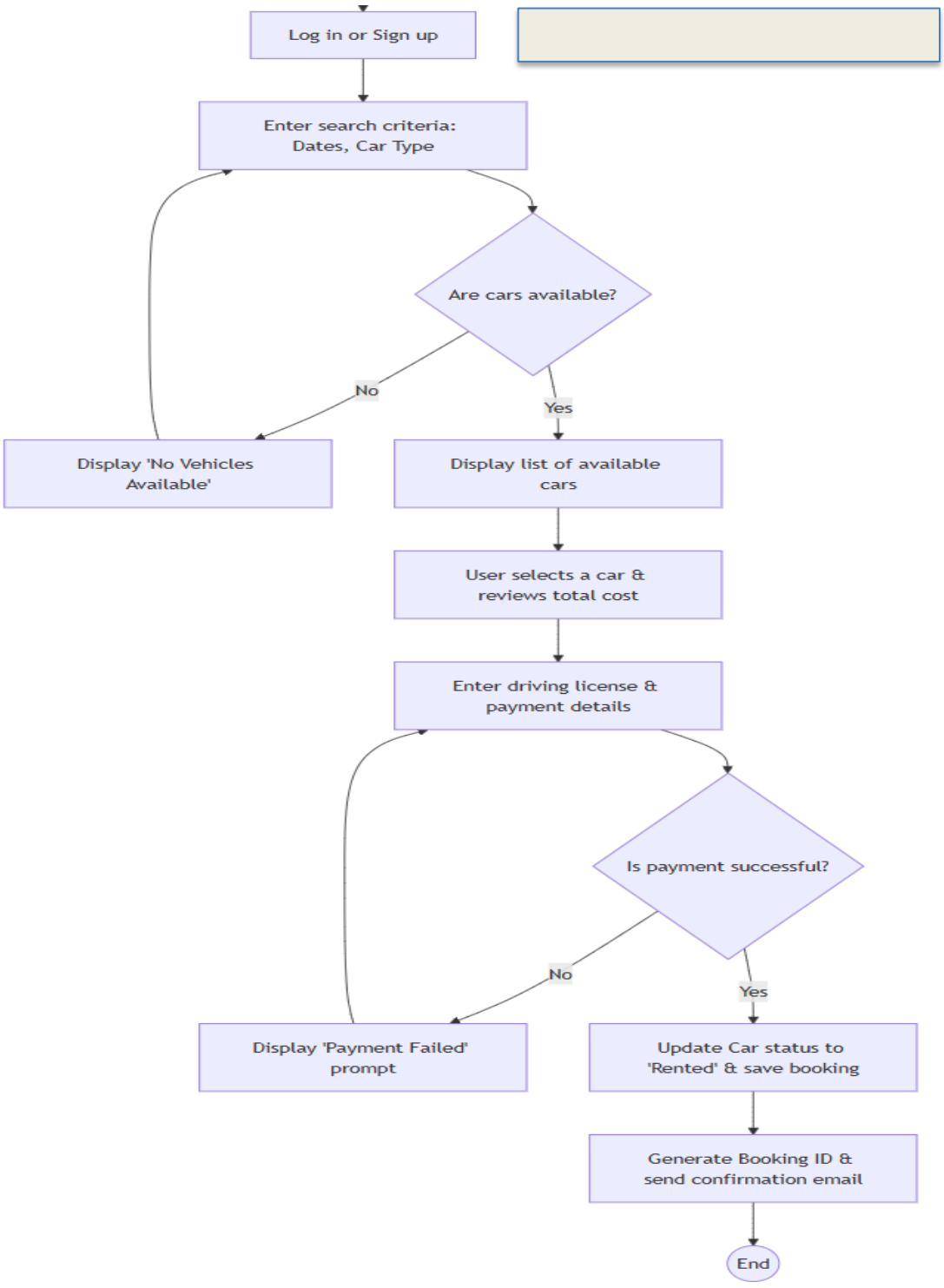
As the former Levels were discussed, the DFD Level0 shows the main concept, the level 1 broadens it then the level 2 placed the database used to store all the data that flows in Car Rental System. Data Flow Diagram level 2 let's you know the ideas on where does the data inputs goes and outputs comes within the project.



System sequence diagram



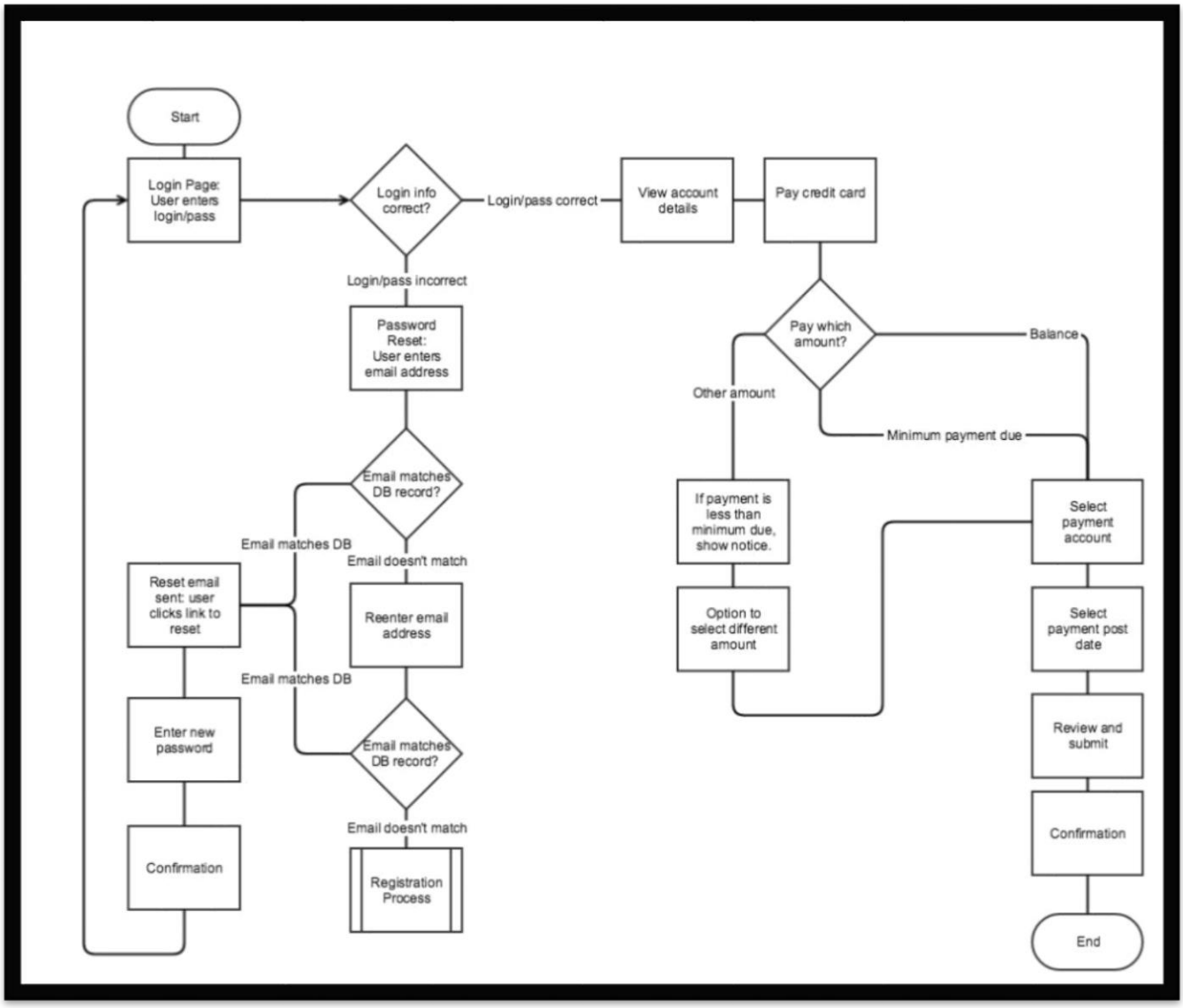
ACTIVITY DIAGRAM



4. System Design & Data Modelling

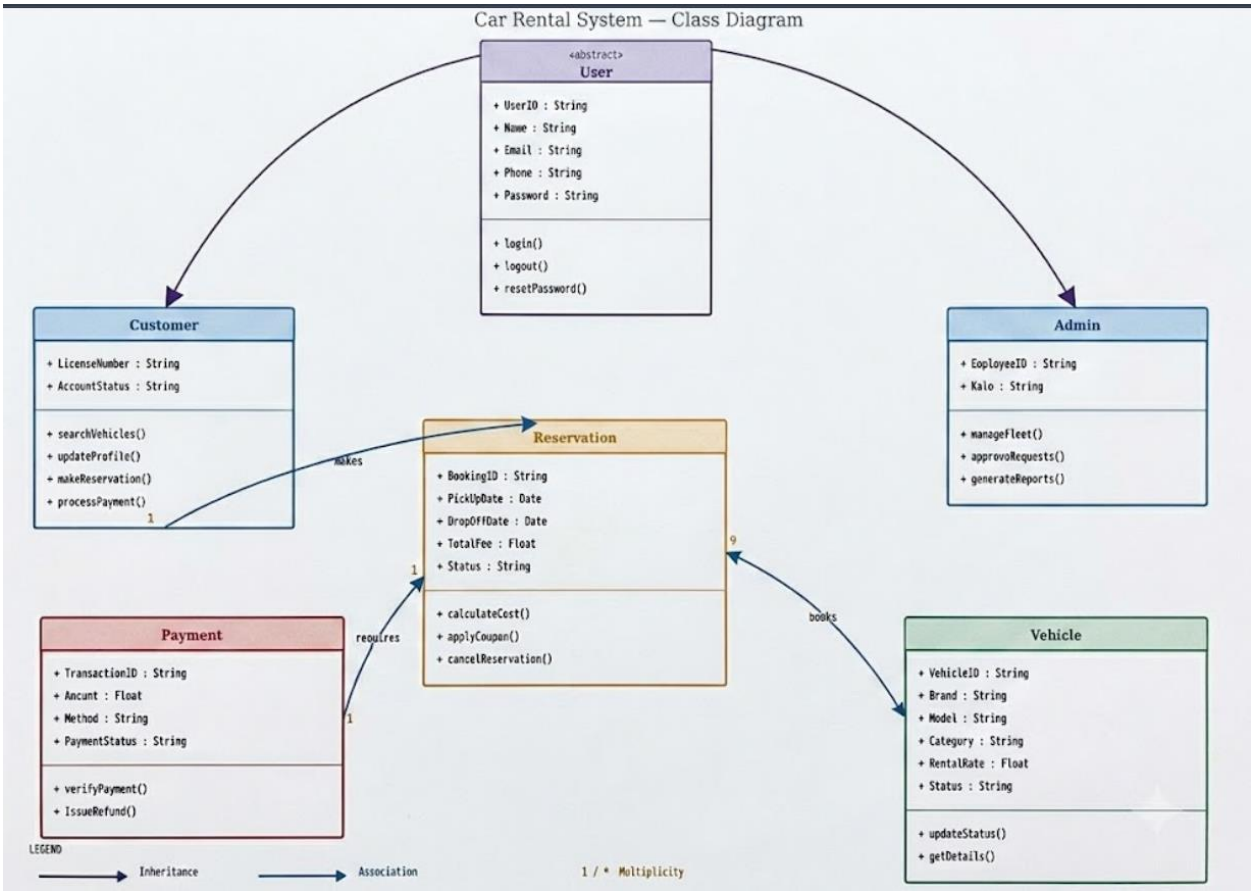
4.1 Entity Relationship Diagram (ERD)

The database is structured to maintain referential integrity. Primary entities include **Users** (linked to Bookings), **Cars** (linked to Bookings and Categories), and **Payments** (linked to Bookings).

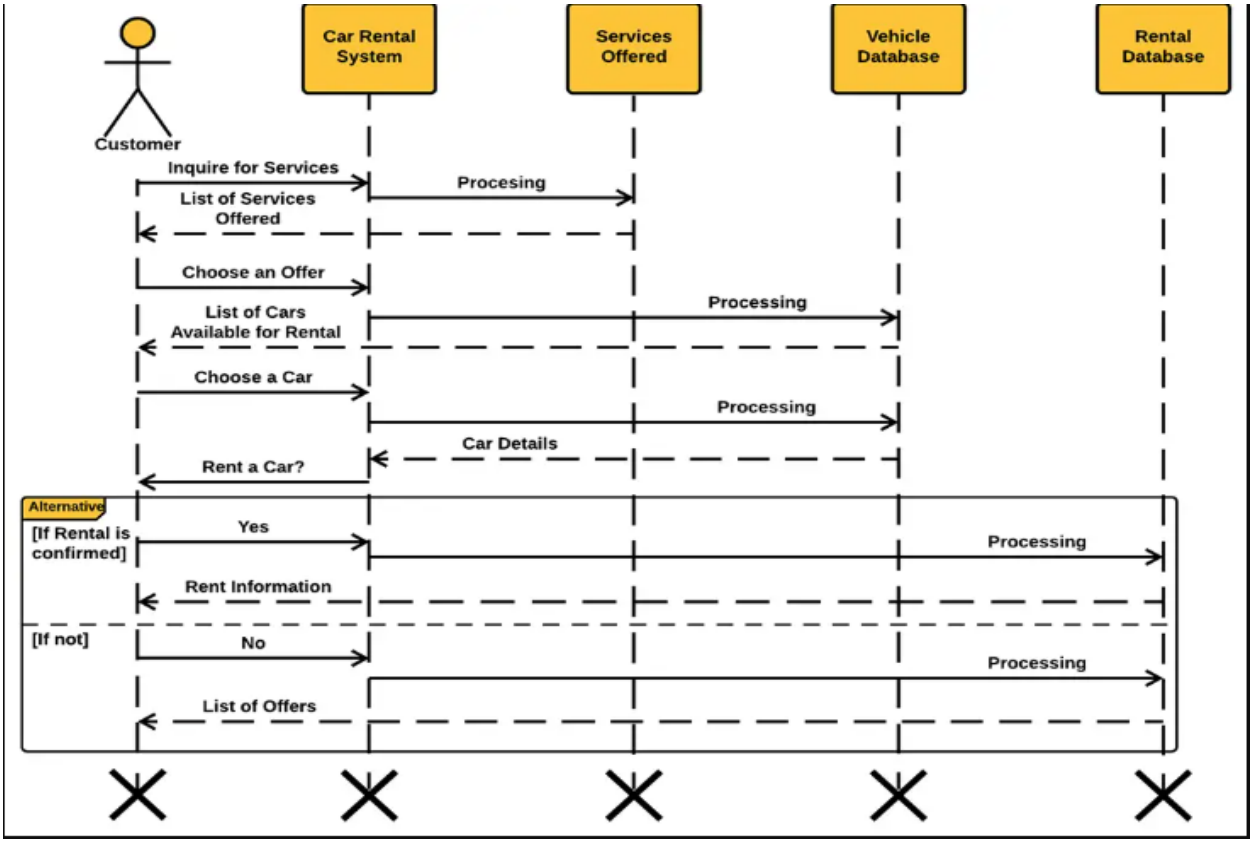


4.2 Class Diagram:

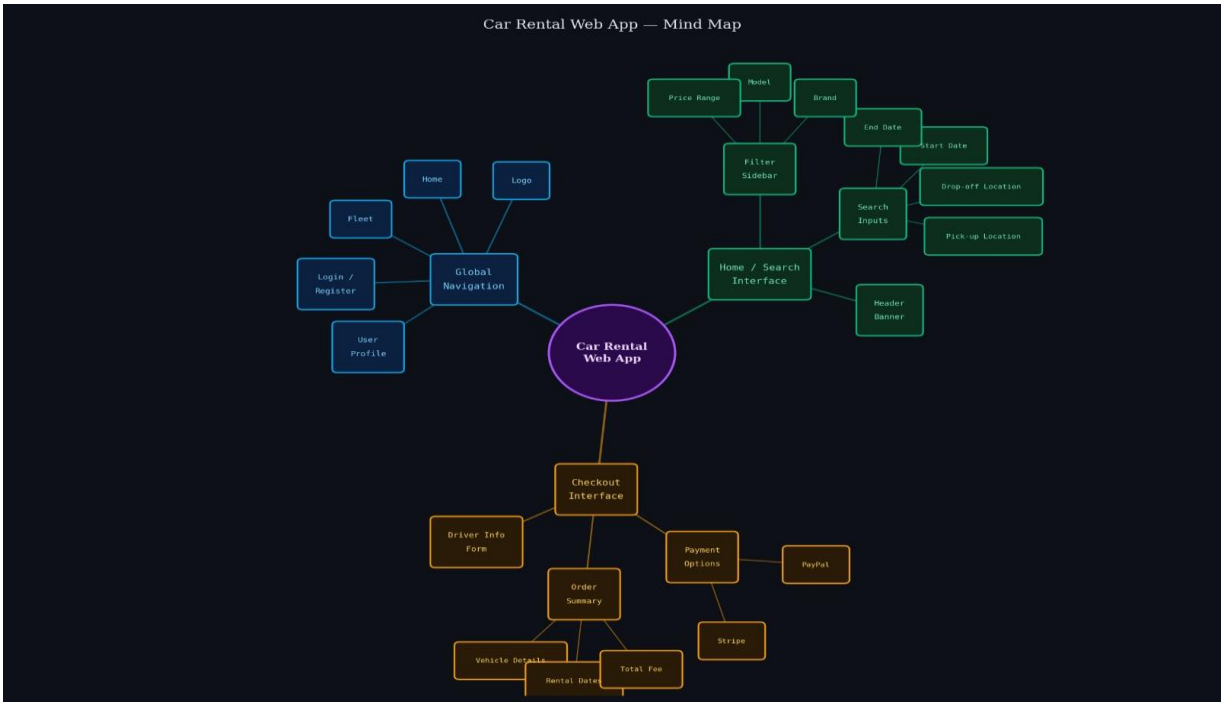
The **MySQL** schema will consist of normalized tables to reduce redundancy. For example, the 'Cars' table will store static data (Model, Plate), while a separate 'Status' table will track real-time availability to optimize query performance.



4.3 Sequence diagram:



4.4Interface diagram



4.5 ARCHITECTURE :

The car rental system is built upon 3 layered architecture :

1. **Front End:**

it handle user interface and replace paperwork with the interfaces.

2. **Backend :**

conatains the buissness logicincludes algorithms ,APIs

3. **Database**

- My sql schema normaliza tables used or databases
- with help of tables we secure and manage data easily.

6. TESTING:

Testing is really important. It makes sure the system works properly and does what the system is supposed to do. The system has to be checked for bugs. Testing makes sure everything works together.

Testing includes a things.

- Unit Testing checks parts like the payment calculator to make sure the payment calculator works right.
- Integration Testing checks that the frontend and the backend and the

database all work together without any problems.

- System Testing checks the system to make sure the system meets all the requirements, like the ones listed in the SRS.
- Testing is necessary to ensure the system is good and meets all the requirements. The system has to be tested to make sure the system is free of bugs and the system works properly. Testing is important, for the system.

6.1 Acceptance testing:

- User Acceptance Testing or UAT is the step in testing.
- It is done by the people who will actually use the system like end users and stakeholders.
- The main aim of UAT is to check if the system's ready to be used in a real car rental situation.
- This testing makes sure that the new software works well and replaces the paper systems.
- Users should be able to search for rentals book them and pay without any problems, like delays or mistakes.
- The system needs to work in a real-world car rental environment.
- UAT ensures that software meets the needs of users in a car rental setting.

6.2 Test Case:

Test Case ID : TC-001 The Test Title is Verify User Registration Process. The Requirement ID is FR001.

To start the test the user must be on the Registration Page. Have a stable internet connection.

Here are the steps to follow for the User Registration Process:

1. The user needs to enter a Name.

2. Then the user has to enter an Email address for the User Registration Process.

3. Next the user has to enter an 11-digit phone number for the User Registration Process.

4. After that the user needs to enter a password for the User Registration Process.

5. Finally the user has to click on the Register button to complete the User Registration Process.

The Expected Result of the User Registration Process is that the system must save the data securely it must create the user profile. It must redirect the user to the Login page with a success message, for the User Registration Process.

The Actual Result of the User Registration Process is to be filled during testing.

The Pass or Fail status of the User Registration Process is also to be filled during testing.

5. References

- IEEE Standard 830-1998 for Software Requirements Specifications.
- Pakwheels
- Rent a car.
- https://www.rentacarservices247.com/